

Do You XaRaoke? Immersive Realistic Singing Experience with Embodied Singer

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ABSTRACT

We have developed an immersive karaoke experience that allows users to sing their favorite songs in front of a simulated audience. The karaoke experience is designed within an immersive stadium scene with a simulated audience and stage lights that synchronize with the song's beats and displayed lyrics. Unlike VR-based karaoke commercial solutions, users can even see their real bodies as video-based self-avatars through the use of a deep learning network and a real microphone without using VR controllers. Preliminary results from a subset of 17 participants validate the developed prototype and provide insights for future improvements.

Index Terms: immersive karaoke —crowd simulation—self-embodiment

1 INTRODUCTION

Karaoke is a form of entertainment in which people sing in front of the public along with recorded music and transcribed lyrics using a microphone. It has become a popular pastime in many countries around the world, especially in Japan and South Korea, and it is often enjoyed in bars, clubs, and private parties. Among its benefits karaoke allows people to express themselves, release emotions and improve mood, as well as being an opportunity for people to connect and have fun together. Sul *et al.* were one of the first ones to develop a virtual stage recreating a karaoke by rendering the image of the singer on top of a non-immersive 3D scene [5], providing interactive visual feedback such as virtual musicians, dynamic stage and lights, among others. Immersive karaoke has been also proposed to enhance language pronunciation [1]. From the commercial side, there are only a few VR products related to the karaoke experience: Sing Together VR, released in 2021 by Stasis Field, and the *Enjoy Japanese Karaoke* new functionality provided by VARK app.

In this work, we have developed an immersive experience that aims to replicate the experience of traditional karaoke. This immersive experience takes place inside a 3D scene of a stadium during a concert full of people and with dynamic lights. Users can choose from a wide range of songs and sing along with the lyrics displayed on the Virtual Reality (VR) headset. A microphone captures the user's singing voice. Besides, to increase the realism of the experience, we incorporate the user's own body into the experience as well as a real microphone.

2 KARAOKE PIPELINE

The entire application has been developed in Unity 2020, using a workstation with an Intel Core i7-11800H, 32GB RAM, and one Nvidia GTX 3070 GPU and Meta Quest 2 as the VR googles. The procedure is as follows:

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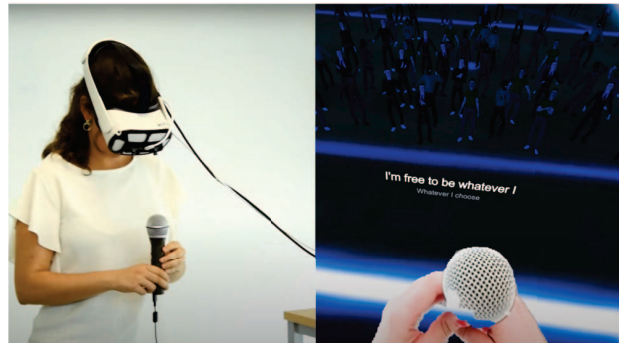


Figure 1: XaRaoke view. Left) A user wearing Oculus Quest 2 with a 3D printed mounted piece to hold a stereo RGB camera; Right) View of the immersive karaoke experience, including the representation of the user's body in the experience.

Given a requested song, we employ the Youtube DATA API to obtain the audio files. Before proceeding, we ensure the existence of the song by utilizing a Spotify C# API Wrapper. The playback of songs is accomplished through the handling of mp3 files and seamlessly integrated into the Unity environment. If only the original version with vocals is accessible, we employ a processing step to extract a vocal-free audio version using Spleeter. Lyrics are loaded through an external free API. The lyrics are converted to an internal data structure that organizes them in timestamped sentences.

Further, the input microphone is sampled at a standard rate of 48KHz. Then, the raw sample is processed using Unity's Fast Fourier Transform to obtain the relative amplitude over a specified number of frequency bins. In this case, we need a detailed enough frequency granularity to be able to distinguish nearby musical notes. Therefore, 2048 bins are used, resulting in 11.71 Hz per bin, which is enough for our purpose. Knowing now the dominant frequency at a given moment in time, we obtain the corresponding musical note through approximation using the standard tuning A4 = 440Hz.

The environment is a karaoke stadium stage with dynamic lights where the user will perform a song in front of a simulated audience. The audience is modeled by placing 3D human models throughout the stadium floor¹. The 3D human model comes with 5 possible animations: a neutral state, 2 clapping styles, and 2 cheering styles. To create a sense of realistic non-verbal feedback from the public, animations are triggered randomly for each modeled person through a Unity script. As an interactivity feature, the public can also provide the user with audio feedback that is driven by the intensity of the user input voice. Aiming to give a more realistic feel, different types of volumetric lights are used. The aim of these light shafts is to

¹<https://www.cgtrader.com/free-3d-models/character/man/audience-unity-ready>

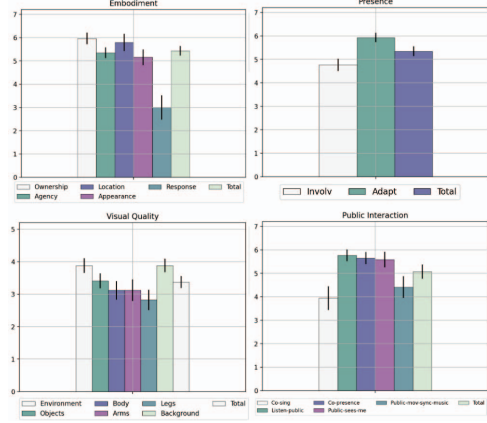


Figure 2: Preliminary Subjective Results. From Up to Down and Left to Right: Embodiment, Presence, Visual Quality, and Public Interaction.

represent stage lights similar to those from a real concert. Lights are animated using DoTween free animation engine. The characteristics of each light, for example, strength or flashing speed, are adapted to each song. These features of the song, mainly its bpm and mood, are retrieved through a Spotify C# API Wrapper.

To increase the realism level of the experience we add the user's own body into the experience. This is done by placing a stereo camera in front of the HMD with the help of a 3D-printed attachment (Fig. 1). The camera captures an egocentric video representing the equivalent view that the user would have with their own eyes if they did not have the HMD on. As a first step in creating the self-avatar, we relied on an algorithm from the work by Gonzalez *et al.* [3] to segment, in real-time, users' own bodies from the egocentric video.

3 PRELIMINARY RESULTS

The experience was tried by 17 subjects (7 men and 9 women, one unknown, ages 42 – 57). We recruit the participants by spreading the word about the experience through an institutional email.

At the beginning, participants were briefed about the experiment and the purpose of it and asked to sign a consent form. Users had the possibility to choose the song they wanted to sing. Once it was finished, they were asked to fill a short Embodiment composed of 13 questions and a set of 6 presence [4] questionnaires, based on Gonzalez-Franco and Peck EQ [2] and Witmer and Singer PQv3 [6], respectively. To measure the visual quality of the user's body and the microphone, a total of 5 extra questions were asked, including one where we asked about how annoying false positives were. An extra question was included to understand whether people experienced sickness or not during the experience. To measure public interaction in the context of the karaoke, a total of 5 questions in 7-grade Likert scale questions were reported (see Table 1). Besides, three additional questions were included to explore people's acceptability.

Table 1: Statements to measure Public Interaction

Factor	Question
Co-singing	<i>I felt that the audience was singing with me</i>
Listening	<i>I felt I was listening to the public</i>
Co-presence	<i>I felt that I was present with the audience on the stage floor</i>
Co-presence	<i>I felt that the audience could see me singing and dancing</i>
Synchronization	<i>I felt the audience move and dance with the rhythm of the song</i>

First, Global Quality of Experience (QoE) was asked using the ITU-T ACR scale. The Net Promoter Score was extracted with two computations: the traditional one, based on the self-reported probability of recommending the game to a friend, ranging from 0 – 10, and a variant based on the self-reported willingness to pay.

Results, displayed in Fig. 2, show that our solution provides high values of avatar embodiment and sense of presence. More concretely, we obtained MOS above 5 for almost all embodiment and presence subfactors. In what concerns the visual quality of the environment and the embodied singer, users perceive their body and microphone with reasonable quality (above 3) but also state that the microphone was not always visible and that the human body was better rendered when the user was not moving. In what concerns, perception of false positives, less than 10% reported to be a little bit annoyance, while the majority reported not to be annoying or even perceivable.

Quality of Experience ratings were relatively high, with above 76% of the participants reporting Good or Better. Net Promoter Scores were also high, mostly in the range of 80%-100%. Also, nearly 60% of the users were willing to pay for this experience.

4 QUALITATIVE FEEDBACK

People also report room for improvement in what concerns rendering between the human body and the 3D environment, as the current status does not look very realistic. Regarding public interactions, high levels of feeling together with the public, and also the feeling of being seen and listened to by the public are achieved. However, users experience less when the public is singing with them or moving with the song's rhythm. This might be probably due to the limited amount of animations available in the 3D human models. Overall, people received the immersive karaoke as a positive experience.

5 CONCLUSIONS

In this work we have developed a XaRaoke: an immersive experience karaoke using extended reality, that brings the possibility to enjoy karaoke without getting out from home, avoiding stage fright. Lyrics are overlaid synchronously with the rhythm of the requested songs. Besides, to enrich the immersion and realism, we integrate the user's own body and microphone into the experience. We carried out preliminary subjective experiments with a subset of 17 people and obtained positive and constructive feedback.

For future work, we plan to extend the functionality of XaRaoke so that it can be used for more than one user, to introduce the option so that the user can freely decide the best place for the lyrics, and the possibility to delete the virtual audience (in case of stage fright).

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